

# Home Tasks - 03

Student Name: Naveen  
 Branch: CSE - AIML  
 Semester: 5  
 Subject Name: ADBMS

UID: 24BAI70397  
 Section/Group: 24AIT\_KRG-1\_A  
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## Q1.

Given a table containing information about bank deposits and withdrawals made using Paypal, write a query to retrieve the final account balance for each account, taking into account all the transactions recorded in the table with the assumption that there are no missing transactions.

### transactions Table:

| Column Name      | Type    |
|------------------|---------|
| transaction_id   | integer |
| account_id       | integer |
| amount           | decimal |
| transaction_type | varchar |

### transactions Example Input:

| transaction_id | account_id | amount | transaction_type |
|----------------|------------|--------|------------------|
| 123            | 101        | 10.00  | Deposit          |
| 124            | 101        | 20.00  | Deposit          |
| 125            | 101        | 5.00   | Withdrawal       |
| 126            | 201        | 20.00  | Deposit          |
| 128            | 201        | 10.00  | Withdrawal       |

### Example Output:

| account_id | final_balance |
|------------|---------------|
| 101        | 25.00         |
| 201        | 10.00         |

**Solution:**

```
SELECT ACCOUNT_ID,  
       SUM (  
         CASE  
           WHEN TRANSACTION_TYPE = 'Deposit' THEN AMOUNT  
           WHEN TRANSACTION_TYPE = 'Withdrawal' THEN -AMOUNT  
         END  
       ) AS FINAL_BALANCE  
FROM TRANSACTIONS  
GROUP BY ACCOUNT_ID;
```